Andrew Marion

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EDUCATION

KENNESAW STATE UNIVERSITY (KSU) | Marietta, GA — Anticipated Graduation May 2027

- *Bachelor of Science in Mechanical Engineering, Aerospace Engineering Minor* — GPA: **3.81/4.0**
- Relevant Coursework: Aerodynamics, Aircraft Design, Fluid Dynamics, Thermodynamics, Heat Transfer, Dynamics, Machine Design, Material Science, Computer Aided Engineering

RESEARCH EXPERIENCE

KSU Advanced Aerodynamics and Propulsion Lab | Marietta, GA — September 2023 – Present

Undergraduate Research Assistant, Advisor: Dr. Gaurav Sharma

Numerical Investigation of Compound Delta Wings — September 2023 – Present

- Executed 140+ RANS CFD simulations in ANSYS Fluent across Mach numbers and angles of attack, increasing lift-to-drag ratio 7.24–13.57% over the baseline wing in the transonic regime
- Applied GPU-accelerated Large Eddy Simulation (LES) to resolve vortex breakdown and validate RANS aerodynamic performance predictions
- Authored and presented first peer-reviewed paper at the 2025 AIAA Region II Student Conference
- Presented findings to the KSU President's Community Advisory Board and the 2024 Symposium of Student Scholars

Multi-Fidelity Assessment of Control Surface Modeling with FlightStream — March 2025 – Present

- Evaluate mid-fidelity control surface deflection modeling in FlightStream to define accuracy boundaries for preliminary aircraft design
- Validated FlightStream predictions against high-fidelity CFD, flight test, and WindShape wind tunnel data; published at AIAA SciTech 2026
- Cross-validated a mid-fidelity fixed-wing aerodynamic model against flight test data for control law development; accepted to AIAA SciTech 2027
- Coordinate data sharing and technical deliverables across a multi-institutional partnership with Oklahoma State University's S2FAR Lab and Siemens

Aerodynamic Characterization of a Blended Wing Body Using FlightStream — January 2026 – Present

- Ran a 42-case transonic RANS sweep in ANSYS Fluent (Mach 0.6–0.8, -5° to 20° angle of attack) to benchmark FlightStream in collaboration with Siemens
- Showed FlightStream solved matched cases $5\times$ faster than RANS with lift coefficient within 0.013 in the linear range, supporting faster early-stage design iteration
- First author on 2 papers accepted to AIAA SciTech 2027; co-author on a Lattice Boltzmann analysis of the same aircraft

PROFESSIONAL EXPERIENCE

NASA Glenn Research Center | Cleveland, OH — August 2026 – December 2026

UAV Flight Control Intern, Diagnostics and Controls Branch

- Implement custom MATLAB/Simulink flight control algorithms on Pixhawk flight computers running PX4 autopilot for a 12 ft wingspan, 16-fan distributed electric propulsion (DEP) UAV based on NASA's SUSAN Electrofan configuration
- Develop fault-tolerant control allocation algorithms in MATLAB that redistribute thrust after fan or thrust-vectoring failures, evaluated across 8,000 Monte Carlo fan-failure cases
- Model UAV flight dynamics and powertrain behavior in MATLAB/Simulink and X-Plane for control law development
- Support thrust-vectoring flight control development for NASA Glenn's Aircraft Propulsion and Power research

Textron GSE | Cartersville, GA — June 2026 – August 2026

Mechanical Engineering Intern, New Product Development

- Supported new product development for the electrification of airport ground support equipment, including tow tractors and belt loaders
- Designed 3 assemblies and 15 parts in Siemens NX with production drawings, including the MX4 electric baggage tractor hinged door and 660 belt loader access doors
- Ran motion simulation and physical prototype tests, including parking brake actuation sensor placement and access door kinematics, verifying function and reducing design rework before production release

KSU Office of Research | Marietta, GA — February 2026 – Present

Research Development Intern

- Conduct funding searches and budget analyses to identify external grant opportunities for KSU faculty
- Design and build web pages communicating funding opportunities and research resources to faculty

TEACHING EXPERIENCE

Kennesaw State University | Marietta, GA — September 2025 – Present

Dynamics Teaching Assistant

- Grade and annotate homework for 80+ students, explaining errors to reinforce dynamics fundamentals
- Hold one-on-one sessions with students on dynamics concepts and problem solving
- Support students over email and Microsoft Teams with homework questions

PUBLICATIONS

Conference Papers

- Marion, A., Sharma, G., Sirak, E., and Shahjahan, S., "Aerodynamic Characterization of a Blended Wing Body Using a Surface Vorticity Solver," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Marion, A., Sharma, G., Sirak, E., and Shahjahan, S., "Modeling Vortex-Dominated Flow over a Delta Wing Using a Surface Vorticity Solver," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Mehar, P., Marion, A., Sharma, G., Muller, M., and Gaedtke, M., "Aerodynamic Analysis of a Blended Wing Body Using the Lattice Boltzmann Approach," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Atkinson, Z., Comer, A., Marion, A., and Sharma, G., "Cross-Validation of a Mid-Fidelity Fixed-Wing Aerodynamic Model for Control Law Development with Flight Testing," accepted for presentation at AIAA SciTech 2027 Forum, January 2027
- Comer, A., Atkinson, Z., Marion, A., Sharma, G., Shahjahan, S., and Wilson, T., "Multi-Fidelity Assessment of Mid-Fidelity Control Surface Modeling with FlightStream," AIAA SciTech 2026 Forum, January 2026. doi.org/10.2514/6.2026-2810
- Marion, A., and Sharma, G., "Numerical Study of Transonic Flow Characteristics Around a Compound Delta Wing," AIAA Region II Student Conference, April 2025. doi.org/10.2514/6.2025-97857

Presentations and Posters

- Marion, A., "Study of Compound Delta Wings and Kennesaw State Undergraduate Research Experience," Presented to the KSU President's Community Advisory Board, May 2025
- Marion, A., and Sharma, G., "Numerical Analysis of Vortex Breakdown Over a Compound Delta Wing," Poster, KSU Symposium of Student Scholars, May 2024

SCHOLARSHIPS AND AWARDS

- Junior Scholars Program, KSU Office of Undergraduate Research ($2,000) — August 2025
- SPCEET Faculty Scholarship Grant for transonic delta wing research under Dr. Gaurav Sharma ($3,000) — Spring 2025
- Dean's List, Kennesaw State University — Spring 2024, Fall 2024, Spring 2025
- First Place, KSU College of Computing and Software Engineering Ideathon ($150) — 2024
- Sophomore Scholars Program, KSU Office of Undergraduate Research ($2,000) — August 2024
- Summer Undergraduate Research Program (SURP), KSU Office of Undergraduate Research ($4,000) — April 2024
- President's List, Kennesaw State University — Fall 2023
- First-Year Scholars Program, KSU Office of Undergraduate Research ($2,000) — August 2023
- Georgia HOPE Scholarship, full undergraduate tuition — May 2023

LEADERSHIP AND COMPETITION TEAMS

Kennesaw Rocketry Team | Marietta, GA — May 2025 – June 2026

Vice President — *December 2025 – June 2026*

- Led the administrative team organizing public outreach events and standardizing documentation while supporting engineering and launch operations
- Organized OpenRocket, SolidWorks, and ANSYS STEM workshops, expanding team technical capabilities and outreach

Liquid Propulsion Lead — *May 2025 – December 2025*

- Designed a 500 N nitrous oxide/ethanol bipropellant liquid rocket engine for future NASA Student Launch entries
- Led propellant selection and injector design trade studies using CAD and CFD

Kennesaw Aerial Robotics Team | Marietta, GA — September 2023 – April 2025

PADA Team Lead

- Led 20+ members through design, manufacturing, and flight testing of a powered autonomous delivery aircraft (PADA) deployed as a parasite aircraft from a larger carrier plane
- Placed 10th of 19 teams in the Advanced Class at the 2024 international SAE Aero Design competition
- Used ArduPilot to fly a custom flying wing autonomously after mid-flight release from the carrier aircraft, navigating to landing coordinates detected by the parent aircraft
- Configured a Matek H743 flight computer and tuned its autopilot for stable flight on an inherently unstable platform

- Taught a three-part ANSYS Fluent workshop series to KSU competition team members

ACADEMIC PROJECTS

Ceramics and Metal Alloys for Re-entry Vehicle Heat Shields – Material Science Spring 2025
- Conducted a literature review with 2 teammates on ceramic and metal alloy heat shields, then wrote a report and presented on their application, manufacturing, and thermal performance

Pumpkin Launch – Dynamics Fall 2024
- Designed and built a 12 ft trebuchet with a 6 ft arm and 200 lb counterweight capable of launching 15 lb pumpkins, using dynamics analysis to size the design

Gravity-Launched Vehicle Design – Introduction to Mechanical Engineering Spring 2024
- Engineered a self-steering vehicle guided by a wire track with consistent results, documented in a technical design report

PROFESSIONAL ORGANIZATIONS

- American Institute of Aeronautics and Astronautics (AIAA), Student Member October 2023 – Present
- American Society of Mechanical Engineers (ASME), Student Member February 2024 – February 2025
- American Physical Society (APS), Student Member October 2023 – October 2024

TECHNICAL SKILLS

CFD and Aerodynamics: ANSYS Fluent and Workbench, FlightStream, SolidWorks Flow Simulation, OpenVSP
Methods: RANS, LES, Transonic Flow, Vortex Lift, Surface Vorticity Methods, Control Allocation, Fault-Tolerant Control, Monte Carlo
Controls and Autonomy: MATLAB/Simulink, PX4, ArduPilot, Pixhawk, Matek H743, X-Plane
Programming: MATLAB, Python, R, LaTeX
CAD: Siemens NX, SolidWorks, AutoCAD, ANSYS SpaceClaim, Engineering Drawings
Manufacturing: 3D Printing, Laser Cutting, Composite Layup, Lathe/Mill, CNC Machining, Foam Cutting
Software: ParaView, Microsoft Office, Inkscape, OriginPro